

ELEC70122: Machine Learning for Safety-Critical Decision-Making

Problem Set 3

Due date: 6 March 2026, 11h59PM
Total Marks: 30

Submission Instructions

Upload a PDF of your solutions to the following problems to Blackboard assignments for marking. Late submissions will be penalized 10% per day late and will NOT be marked if delayed by more than 3 days. Should you have problems with uploading, please contact the TAs listed on Blackboard.

1 Causal Inference

1.1 Expressing Causal Problems in Math

In this problem, we will present you with several free text scenarios. For each scenario, you must answer first whether or not causal inference is required in this scenario, and, if so, you must identify the relevant covariates (X), treatments (T), outcomes (Y), and any hidden confounders (H) that pose particular concern in this setting. Learning Goals: We hope that this problem will help you develop the ability to recognize causal problems in the wild, to be able to translate them into the mathematical framework discussed in lectures, and to think critically about any missing confounders.

1. You have collected physiological timeseries data (in particular, EEG traces, heart rate, SpO2, and blood pressure) from patients during a set of surgical procedures which are controlled by the application of anesthetic drug D at various dose levels continuously throughout the procedure. While most patients survive the procedure, a small minority do not. You want to design a model to automatically control the drug dosage level over time to maximize chance of patient survival. (4 marks)
2. You work for a hospital and have just been assigned to help a new payer (e.g., insurance agent) satisfy an odd request. They like to receive advanced estimates of billing codes they will eventually be charged for their patients, and have requested that you provide them with an estimate of each patient's final billing code after only their first 24 hours. Contractually, this payer is prohibited from using these estimates to affect whether they accept or dispute claims, but instead uses them to more proactively assess their revenue streams. To accommodate this request, you plan to build a model based on retrospective data to take the first 24 hours of a patient's stay and predict the final ICD10 codes they received at discharge time. (4 marks)
3. You work for a hospital which, like many hospitals, receives penalties from major payers if patients are readmitted within 30 days of discharge. To help prevent this, you decide to train a model based on past patient data, ingesting all EHR data up to discharge and predicting whether or not the patient will be readmitted within 30 days. Ultimately, you hope to use this model to audit and inform discharge decisions. (4 marks)

1.2 Calculating ATE and CATE

1. The data in Table 1 comes from an experiment run from 2004-2008 at Sunnydale Hospital. In the data, you have binary indicators for: prior education (Z), whether the surgeon had ≥ 100 successful surgeries in 2004 (X), whether the surgeon enrolled in the fellowship from 2005-2006 (T), whether the surgeon had ≥ 150

successful surgeries in 2007 (Y), and whether the surgeon cumulatively had ≥ 500 successful surgeries in their lifetime by 2008 (W). We know the following: (i) The number of successful surgeries in 2004 depends solely on the prior education. (ii) Whether a doctor is selected to the surgery fellowship program depends on their prior education and number of successful surgeries in 2004. (iii) The number of successful surgeries in 2007 depends on the fellowship, prior education, and number of successful surgeries in 2004. (iv) The cumulative number of successful surgeries by 2008 is directly based on how many successful surgeries a doctor performs in 2004 and 2007. Draw the causal graph that describes the above experiment. (4 marks)

Z	X	T	Y	W
0	1	1	1	0
0	0	1	1	0
0	0	1	1	0
1	1	1	1	1
1	0	1	1	1
1	0	1	1	1
1	1	1	0	0
1	0	1	0	0
0	0	1	0	0
1	1	0	1	1
1	0	0	1	0
1	0	0	1	0
1	1	0	0	1
1	1	0	0	1
0	1	0	0	0
1	0	0	0	0
0	0	0	0	0
0	0	0	0	0

Table 1: Sample Data for Problem 1.2

2. Calculate the Average Treatment Effect (ATE) of the fellowship (T) on 2007 success (Y). Use covariate adjustment and empirically estimate the probabilities/expectations from the observed data. Recall that

$$ATE := \mathbb{E}[Y_1 - Y_0], \quad (1)$$

where for this causal graph,

$$\mathbb{E}[Y_t] = \sum_{z \in \{0,1\}} \sum_{x \in \{0,1\}} P(Z = z)P(X = x|Z = z)\mathbb{E}[Y|X = x, Z = z, T = t] \quad (2)$$

(4 marks)

3. Calculate the Conditional Average Treatment Effect (CATE) of the fellowship (T) on 2007 success (Y) for patients without prior education ($Z = 0$). Recall that,

$$CATE := \mathbb{E}[Y_1 - Y_0|Z = 0] = \mathbb{E}[Y_1|Z = 0] - \mathbb{E}[Y_0|Z = 0], \quad (3)$$

where for this causal graph,

$$\mathbb{E}[Y_t|Z = 0] = \sum_{x \in \{0,1\}} P(X = x|Z = 0)\mathbb{E}[Y|X = x, Z = 0, T = t] \quad (4)$$

(4 marks)

2 Reinforcement Learning

Off-policy evaluation (OPE) aims to estimate the expected return of a policy using data collected from a different policy. Causal inference methods can help improve OPE by estimating counterfactual outcomes under different interventions.

1. Explain how causal inference can be used to address distributional mismatch in OPE. (2 marks)
2. Consider a setting where an agent interacts with an environment and collects data following a behavior policy π_b . Describe how you could use inverse propensity scoring (IPS) or a related method to estimate the value of a target policy π_e . What assumptions are necessary for this approach to yield an unbiased estimate? (2 marks)
3. Discuss one challenge that arises when using causal graphical models for OPE in real-world applications, and suggest a potential solution. (2 marks)